

Lec 5. System of 1st order linear Eqn.

(I) Motivation: General n -th order constant coefficient

linear ODE: $y = y(t)$

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_0y = f(t)$$

Introduce:

$$x_1(t) = y$$

$$x_2(t) = y' \Rightarrow \frac{d}{dt} x_i(t) = x_{i+1}, \quad i=0, 1, \dots, n-1$$

$\vdots$

$$x_n(t) = y^{(n-1)}$$

$$\begin{aligned} \frac{d}{dt} x_n &= \frac{d}{dt} y^{(n-1)} = \frac{d^n y}{dt^n} = -[a_{n-1}y^{(n-1)} + \dots + a_0y] \\ &= -a_{n-1}x_n - \dots - a_0x_1 + f(t) \end{aligned}$$

$$\frac{d}{dt} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{pmatrix} & & 1 \\ & & \\ & & \\ -a_0 & \dots & -a_{n-1} \end{pmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ f(t) \end{bmatrix}$$

$$\text{E.g. } y'' + py' + qy = f(t)$$

(II) Basic Concept.

General: 1st order system

$$\frac{d}{dt} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{pmatrix} F_1(t, x_1, \dots, x_n) \\ \vdots \\ F_n(t, x_1, \dots, x_n) \end{pmatrix}$$

[We shall discuss existence and uniqueness of solution later]

Initial condition

$$x_i(t_0) = x_i^0, \quad \dots, \quad x_n(t_0) = x_n^0$$

Linear: If F_i is linear in unknowns $x_1, \dots, x_n$.

Non linear: otherwise.

$$F_i(t, x, \dots, x_n) = a_{i1} x_1 + a_{i2} x_2 + \dots + a_{in}(t) x_n(t) + b_i(t)$$

Matrix Form :

$$X(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}, A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix}, b = \begin{pmatrix} b_1(t) \\ \vdots \\ b_n(t) \end{pmatrix}$$

Then get

$$\frac{d}{dt} \vec{X}(t) = A(t) \vec{X}(t) + \vec{b}(t)$$

Homogeneous: $\vec{b}(t) = 0$

Nonhomogeneous: $\vec{b}(t) \neq 0$

Existence and Uniqueness Thm. (To be proved).
[Thm 7-1.1, 7-1.2]. Existence and uniqueness

(iii): Structure of solution.

First focus on homogeneous eqn:

$$\frac{d}{dt} \vec{X}(t) = A \cdot \vec{X}(t). \quad (1)$$

[BDM. Thm 7-4.1]

Thm: [Linearity] If $X^{(1)}(t), X^{(2)}(t)$ solves ①,
then $\forall$ constants C_1, C_2 ,

$$C_1 X^{(1)}(t) + C_2 X^{(2)}(t) \text{ solves } \textcircled{1}$$

Generalization: $X^{(i)}(t)$ solves ①, $1 \leq i \leq k$, then
 $\sum_{i=1}^k C_i X^{(i)}(t)$ also solves ①

$$W[X^{(1)} \dots X^{(n)}](t) = \det[X^{(1)} \dots X^{(n)}(t)]$$

$$X^{(1)} = \begin{pmatrix} x_{11} \\ \vdots \\ x_{n1} \end{pmatrix}, \dots, X^{(n)}(t) = \begin{pmatrix} x_{1n} \\ \vdots \\ x_{nn} \end{pmatrix}$$

Linearly independent solutions if $W(t) \neq 0, \forall t$.

Lemma: If $\frac{d}{dt} X^{(i)}(t) = A(t) X^{(i)}(t), 1 \leq i \leq n$.

then $\frac{d}{dt} W(t) = \text{tr}[A(t)] \cdot W(t)$.

Pf: See the class. $\Rightarrow$

$$W(t) = W(t_0) \exp\left[\int_{t_0}^t \text{tr}[A(s)] ds\right]$$

$X^{(1)}(t) \dots X^{(n)}(t)$
is called
a fundamental
set of solution

Thm [7.4.3]

Thm: If $W(t_0) \neq 0$, then $W(t) \neq 0 \forall t$:

Q: How does general solution looks like?

Thm [7.4.2] If $x^{(1)} \dots x^{(n)}$ are linearly independent solution of (1)

$$\frac{d}{dt} X(t) = A(t) X(t) \quad (1)$$

then any solution to (1) can be expressed as

$$\phi(t) = \sum_{i=1}^n C_i x^{(i)}(t)$$

Pf: Given a solution $\phi(t)$ to (1), since $W(t_0) \neq 0$ solves.

$$\underbrace{\begin{bmatrix} x^{(1)}(t_0) & \dots & x^{(n)}(t_0) \end{bmatrix}}_{\text{Invertible}} \begin{bmatrix} C_1 \\ \vdots \\ C_n \end{bmatrix} = \phi(t_0)$$

then define

$$\psi(t) = \sum C_i x^{(i)}(t)$$

Both ϕ, ψ solves (1) with $\phi(t_0) = \psi(t_0)$.

By uniqueness thm $\phi \equiv \psi$. QED

Remark: Solution space is n -D

Q: How to get linearly independent soln?

Thm [7.4.4] Let $x^{(i)}(t)$ be the unique solution to

$$\frac{d}{dt} x^{(i)}(t) = A(t) x^{(i)}(t)$$

$$x^{(i)}(t) = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \rightarrow i\text{-th components.}$$

Then

$x^{(1)}(t), x^{(2)}(t), \dots, x^{(n)}(t)$ form a fundamental set of solution

(IV): Homogeneous linear system with constant coeffs.

• When $A(t) \equiv A$ we can fully solve $x^{(1)}, \dots, x^{(n)}$.

• $\frac{d}{dt} x(t) = A \cdot x(t), \quad A \in \mathbb{R}^{n \times n}, \quad x(t) \in \mathbb{R}^{n \times 1}$ ①

• motivate in 1D:

$$\frac{d}{dt} x(t) = a \cdot x(t), \Rightarrow x(t) = e^{at} x_0$$

• Try

$$x(t) = e^{\lambda t} \cdot v, \quad v \in \mathbb{R}^{n \times 1}$$

⇒ ① ⇒ $\lambda \cdot e^{\lambda t} \cdot v = A \cdot [e^{\lambda t} v] = e^{\lambda t} \cdot A v$

$$\Leftrightarrow \lambda v = A v$$

⇒ (λ, v) is an eigen-pair of A .

Want to find n solution pairs, ideally real.

Few possibilities

1. all real eigenvalues and are distinct. 😊 ✓
2. Some eigenvalues complex.
 ↓ meaning of this case.
3. repeated.

1. 😊 $x(t) = \sum c_i e^{\lambda_i t} v_i$

(IV). a. complex root

$$\text{E.g. } \frac{d}{dt} x = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} x = A \cdot x$$

$$\lambda_1 = i, \lambda_2 = -i$$

$$0 = (A - iI)v_1 = \begin{pmatrix} -i & 1 \\ -1 & -i \end{pmatrix} v_1, v_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$0 = (A + iI)v_2 = \begin{pmatrix} i & 1 \\ -1 & i \end{pmatrix} v_2, v_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

• complex pair, α complex.

$$\begin{cases} AV = \alpha V & \alpha = \lambda + i\mu \\ A\bar{V} = \bar{\alpha}\bar{V} & V = a + ib \end{cases}$$

$$\Rightarrow A(a + ib) = (\lambda + i\mu)(a + ib)$$

$$\begin{aligned} e^{\alpha t} V &= e^{(\lambda + i\mu)t} \cdot (a + ib) \\ &= e^{\lambda t} [\cos \mu t + i \sin \mu t] (a + ib) \\ &= e^{\lambda t} [\cos \mu t \cdot \vec{a} - \sin \mu t \cdot \vec{b} \\ &\quad + (a \sin \mu t + b \cos \mu t) i] \\ &= \vec{p}(t) + i \vec{q}(t). \end{aligned}$$

$$e^{\bar{\alpha} t} \bar{V} = \vec{p}(t) - i \vec{q}(t).$$

can find 2 real solution from

$$c_1 e^{\alpha t} V + c_2 e^{\bar{\alpha} t} \bar{V}.$$

$$\lambda_2 = \bar{\lambda}_1.$$

General solution rewrite $C_1 \cdot e^{\lambda_1 t} V + C_2 \cdot e^{\lambda_2 t} \bar{V}$

E.g. $\frac{d}{dt} X = A X, \quad A = \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}$

meaning of the solution. Do this example.

(IV).b. Repeated root. Tricky

We have generalized eigenvector.

Leave general case to exponential / Jordan normal form

Case I: A has n linear independent eigenvectors (λ_i, V_i)
 $\hat{X} = A X, \quad \lambda_1 = \lambda_2$

While $\lambda_1 = \lambda_2$, still get

$X^{(1)} = e^{\lambda_1 t} V_1, \quad X^{(2)} = e^{\lambda_2 t} V_2$
 and are different.

A is diagonalizable.

Thm: If $A \in \mathbb{R}^{n \times n}$ is symmetric, i.e. $A^T = A$

$\exists$ orthogonal matrix s.t

$A = Q \Lambda Q^{-1}$, Λ diagonal matrix
with real eigenvalues λ_i

[linear algebra]

$\Rightarrow$

$$\frac{d}{dt} x = Ax, \text{ if } A = A^T \Rightarrow A v_i = \lambda_i v_i$$

$$x(t) = \sum C_i e^{\lambda_i t} v_i.$$

Case II:

Consider

$$\frac{d}{dt} X(t) = \begin{pmatrix} \lambda & 1 & \\ & \lambda & \\ & & \lambda \end{pmatrix} X(t), \quad X = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix}$$

Yield

$$\lambda_1(t) = \lambda_2(t) = \lambda_3(t).$$

observe

$$\frac{d}{dt} x_3(t) = \lambda x_3(t) \Rightarrow x_3(t) = e^{\lambda t} x_3(0)$$

$$\frac{d}{dt} x_2(t) = \lambda x_2(t) + x_3(t)$$

$$= \lambda x_2(t) + e^{\lambda t} x_3(0), \quad \Rightarrow x_2(t) \\ \Rightarrow x_1(t).$$

$$\frac{d}{dt} x_1(t) = \lambda x_1(t) + x_2(t).$$